Encoding Auditory Space in Marmoset Cortex

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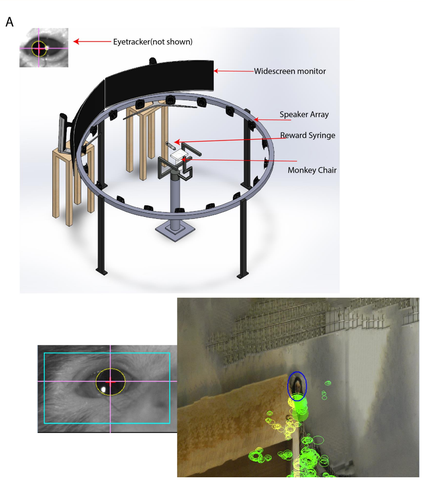
ABSTRACT

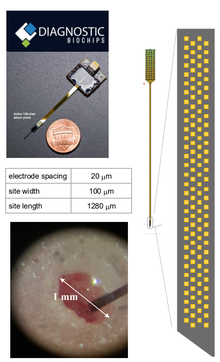
A century ago Jakob von Uexküll argued that every species inhabits its own *Umwelt* – a subjective perception of the physical world shaped by its sensory and motor capacities. For foveate primates such as the common marmoset, vision captures only the frontal space, leaving the vast rear hemisphere to be monitored acoustically. How does auditory cortex meet this ecological challenge, and how is its code influenced by visual events that compete for attention in the frontal field? The results from this study suggest a dual strategy for constructing a 360° auditory *Umwelt*: stable directional "anchors" anchored by contralateral sharp-tuned neurons, and dynamic gain control that boosts regions left outside the visual fovea when frontal space becomes behaviorally loaded.

Our findings bridge classic spatial-hearing theory with *Umwelt* thinking and offer a neural explanation for how primates maintain situational awareness while looking forward.

APPROACH



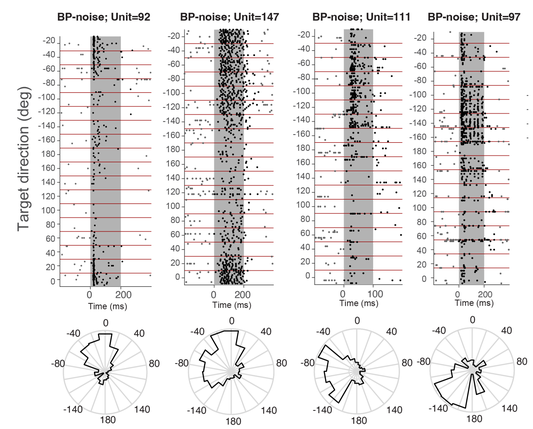
128-channel silicon neural probe



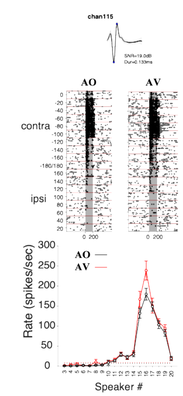
- Head-fixed; no auditory tasks
- SRF based on responses to broadband noise stimuli (200 msec) spanning 360 degree with 20 degree spacing
- Wide-field visual image (±40° azimuth) was presented at 0°
- Single-unit activities collected from the auditory cortex in four marmosets
- 128-chan silicon probes
- Infra-red eye tracking system synchronized with TDT neurophysiology setup

RESULTS

Spatial tuning of ACx neurons across 360°

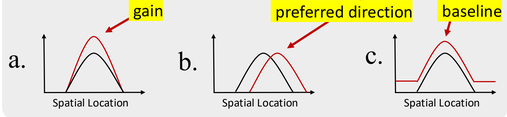


Visual influence on spatial tuning of ACx neurons



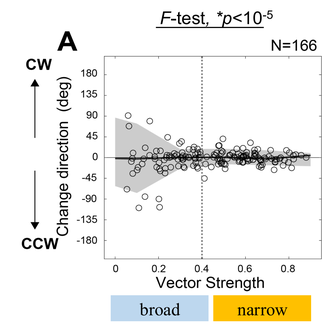
$$Rate_{AV}(\theta) = k_{AV} * Rate_{AO}(\theta - \theta_{AV}) + b_{AV}$$

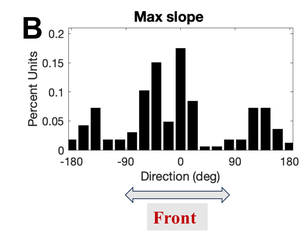
gain | preferred direction | baseline

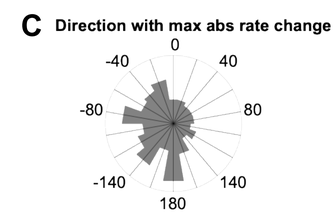


Linear regression model ($p<0.05$ for 155/166 units)

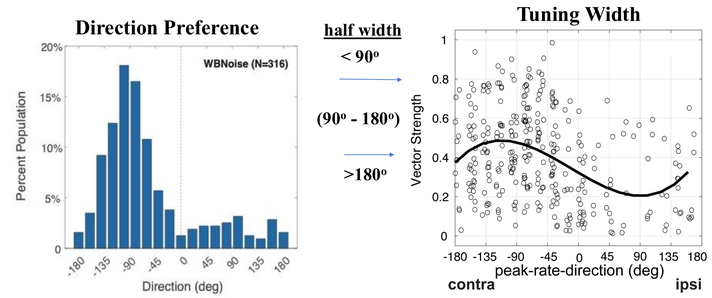
- Quartiles of k [0.67 0.88 1.13]
- Quartiles of b [0.02 0.05 0.10]
- Quartiles of R^2 [0.53 0.73 0.86]



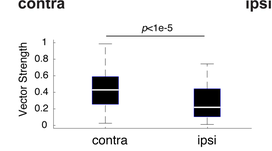




Direction–sharpness coupling



Tuning sharpened progressively toward the contralateral pole, culminating in a dense cluster of narrowly tuned neurons at –90°.



A. Peaking tuning direction does not change with frontal vision, especially for narrowly tuned neurons.
B. The highest spatial sensitivity remains in the frontal and central field, exactly where gaze can be rapidly deployed.
C. Discharge rates changed more at the rear directions, suggesting marmosets maintain situational awareness while looking forward.

SUMMARY

(1) **Direction–sharpness coupling.** Sharply tuned neurons preferred contralateral direction, centered around -90° across the interaural axis.

(2) **Vision-dependent gain without remapping.** The frontal wide-field image multiplicatively scaled firing rates yet left best azimuth unchanged, revealing a cross-modal gain field rather than a spatial realignment.

(3) **Rear-space enhancement.** Gain modulation of firing rates was strongest in the contra-rear field, suggesting that visual stimulation in front strengthens monitoring of the invisible rear hemifield.

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